

MySimulation_19

Simulation File: C:\Users\wotjd\Downloads\Refrigeration project.dwxmz
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CD-1 (Material Stream Cooler)

Property	Value	Unit
Flow Conductance	1	[kg/s]/[Pa^0.5]
Volume	1	m3
Minimum Pressure	101325	Pa
Initialize using Inlet Stream	-1	
Reset Content	0	
Ambient Temperature	298.15	K
Ambient UA Product	0	W/[m2.K]
Wall Thermal Mass	0	
Wall Temperature	298.15	K
Pressure Drop	0	Pa
Efficiency	100	
Outlet Temperature	305	K
Heat Removed	15.9372	kW
Outlet Molar Vapor Fraction	0	
Temperature Difference	-20.138	K.

EV-1 (Material Stream Heater)

Property	Value	Unit
Flow Conductance	1	[kg/s]/[Pa^0.5]
Volume	1	m3
Minimum Pressure	101325	Pa
Initialize using Inlet Stream	-1	
Reset Content	0	
Ambient Temperature	298.15	K
Ambient UA Product	0	W/[m2.K]
Wall Thermal Mass	0	
Wall Temperature	298.15	K
Pressure Drop	0	Pa
Efficiency	100	
Outlet Temperature	253	K
Heat Added	11.7348	kW
Outlet Molar Vapor Fraction	1	
Temperature Difference	0.00542038	K.

C-1 (Adiabatic Compressor)

Property	Value	Unit
Flow Conductance	1	[kg/s]/[Pa^0.5]
Volume	0.01	m3
Minimum Pressure	101325	Pa
Initialize using Inlet Stream	-1	
Reset Content	0	
Rotational Inertia	0	
Current Speed	3000	
Target Speed	3000	
Motor Torque	200	
Surge Flow Fraction	0	
Surge Alarm	0	
Pressure Increase	1.005E+06	Pa
Adiabatic Efficiency	80	%
Temperature Difference	72.1434	K.
Power Required	4.20275	kW
Outlet Pressure	1.25E+06	Pa
PolytropicEfficiency	83.9906	%
AdiabaticCoefficient	1.01398	
PolytropicCoefficient	1.07856	
AdiabaticHead	7875.74	m
PolytropicHead	8268.6	m
RotationSpeed	1500	rpm
PressureRatio	5.10204	

V-1 (Adiabatic Valve)

Property	Value	Unit
Actuator Delay	0	s
Actuator Time Constant	0	s
Opening Setpoint	100	
Cavitation Alarm	0	
Liquid Pressure Recovery Factor FL	0.9	
Calculation Mode	1	
Pressure Drop	1.005E+06	Pa
Outlet Pressure	245000	Pa
Temperature Drop	-52.0054	K.
Kv(max)	100	
Opening	50	
Characteristic Parameter	50	
Actual Flow Coefficient	100	

W_comp (Energy Stream)

Property	Value	Unit
Energy Flow	4.20275	kW

Q_cond (Energy Stream)

Property	Value	Unit
Energy Flow	15.9372	kW

Q_evap (Energy Stream)

Property	Value	Unit
Energy Flow	11.7348	kW

1_evap_out (Material Stream)

Property	Value	Unit
Temperature	252.995	K
Pressure	245000	Pa
Mass Flow	0.0440956	kg/s
Molar Flow	1	mol/s
Volumetric Flow	0.00802833	m3/s
Density (Mixture)	5.4925	kg/m3
Molecular Weight (Mixture)	44.0956	kg/kmol
Specific Enthalpy (Mixture)	-80.0775	kJ/kg
Specific Entropy (Mixture)	-0.447798	kJ/[kg.K]
Molar Enthalpy (Mixture)	-3531.07	kJ/kmol
Molar Entropy (Mixture)	-19.7459	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0.0135163	W/[m.K]

2_comp_out (Material Stream)

Property	Value	Unit
Temperature	325.138	K
Pressure	1.25E+06	Pa
Mass Flow	0.0440956	kg/s
Molar Flow	1	mol/s
Volumetric Flow	0.00177187	m3/s
Density (Mixture)	24.8865	kg/m3
Molecular Weight (Mixture)	44.0956	kg/kmol
Specific Enthalpy (Mixture)	15.2325	kJ/kg
Specific Entropy (Mixture)	-0.388308	kJ/[kg.K]
Molar Enthalpy (Mixture)	671.685	kJ/kmol
Molar Entropy (Mixture)	-17.1227	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0.0225263	W/[m.K]

3_cond_out (Material Stream)

Property	Value	Unit
Temperature	305	K
Pressure	1.25E+06	Pa
Mass Flow	0.0440956	kg/s
Molar Flow	1	mol/s
Volumetric Flow	9.14128E-05	m3/s
Density (Mixture)	482.379	kg/m3
Molecular Weight (Mixture)	44.0956	kg/kmol
Specific Enthalpy (Mixture)	-346.191	kJ/kg
Specific Entropy (Mixture)	-1.55496	kJ/[kg.K]
Molar Enthalpy (Mixture)	-15265.5	kJ/kmol
Molar Entropy (Mixture)	-68.5671	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0.0894887	W/[m.K]

4_valve_out (Material Stream)

Property	Value	Unit
Temperature	252.995	K
Pressure	245000	Pa
Mass Flow	0.0440956	kg/s
Molar Flow	1	mol/s
Volumetric Flow	0.00275979	m3/s
Density (Mixture)	15.9779	kg/m3
Molecular Weight (Mixture)	44.0956	kg/kmol
Specific Enthalpy (Mixture)	-346.191	kJ/kg
Specific Entropy (Mixture)	-1.49871	kJ/[kg.K]
Molar Enthalpy (Mixture)	-15265.5	kJ/kmol
Molar Entropy (Mixture)	-66.0866	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0.0822715	W/[m.K]

5_evap_out (Material Stream)

Property	Value	Unit
Temperature	253	K
Pressure	245000	Pa
Mass Flow	0.0440956	kg/s
Molar Flow	1	mol/s
Volumetric Flow	0.00802853	m3/s
Density (Mixture)	5.49236	kg/m3
Molecular Weight (Mixture)	44.0956	kg/kmol
Specific Enthalpy (Mixture)	-80.0691	kJ/kg
Specific Entropy (Mixture)	-0.447765	kJ/[kg.K]
Molar Enthalpy (Mixture)	-3530.7	kJ/kmol
Molar Entropy (Mixture)	-19.7445	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0.0135168	W/[m.K]